

PYTHON PROJECT REPORT

HIMALAYA INVENTORY MANAGEMENT SYSTEM

A console-based product inventory and record management project

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Project Type	Python Programming Project
Company / Case	Himalaya Company Private Limited

Submitted as a record of the completed project and its demonstrated outputs.

1. Introduction

The Himalaya Inventory Management System is a menu-driven console application designed to manage product records. The supplied execution screenshots demonstrate the main inventory operations through a simple text-based interface. The system presents a company menu, accepts a user's choice, performs the selected operation, and displays the resulting product records.

The screenshots show product records containing a product name, product number, number of units in stock, number of products ordered, MRP, selling rate, manufacture date and expiry date. The program also demonstrates adding, displaying, searching, deleting and updating records.

2. Objectives

1. Maintain product information in an organized record format.
2. Provide a simple menu through which different inventory operations can be selected.
3. Allow new product details to be added to the stock records.
4. Display existing product records and search for a particular product.
5. Delete unwanted product records using a product name or product number.
6. Update individual fields such as name, product number, stock, ordered quantity, MRP and selling rate.
7. Store and display manufacture and expiry dates along with the product information.

3. Main Menu and Functionalities

Option	Function shown in the supplied output
1	Display product record from the company's stock
2	Add product record to the company's stock
3	Delete product record from the company's stock
4	Update product record in the company's stock
5	Sort product record from the company's stock
6	Give order for product
7	See order history
8	See order invoices
9	Cancel order
10	Delete invoices
11	Exit

The first screen in the evidence clearly lists these eleven top-level choices. The remaining screenshots focus mainly on options 1–4 and therefore form the main testing evidence included in this report.

4. Product Record Structure

The displayed records follow a consistent sequence of fields. A typical record shown in the screenshots is represented as a tuple containing the following information:

Field	Meaning
Product Name	Name used to identify the product.
Product Number	Numeric identifier assigned to the product.
Units in Stock	Current quantity available in stock.
Products Ordered	Number of units ordered for the product.
MRP	Maximum Retail Price shown for the product.
Selling Rate	Selling price/rate of the product.
Manufacture Date	Date on which the product was manufactured.
Expiry Date	Date until which the product is valid.

Examples visible in the evidence include Neem Facewash, Neem Scrub, Baby Soap, Baby Powder, Baby Cream, Body Lotion, Moisturising Lotion and Walnut Scrub. The output also shows dates in Python date-object form, such as `datetime.date(2024, 11, 1)`.

5. Working of the System

Step 1 – Start: The application displays the Himalaya company welcome screen and the main menu.

Step 2 – Select an operation: The user enters a menu choice such as 1 for display, 2 for add, 3 for delete or 4 for update.

Step 3 – Enter details: Depending on the selected operation, the program asks for the relevant product information.

Step 4 – Process the record: The selected product record is displayed and the requested action is performed.

Step 5 – Confirmation: The output reports the result using messages such as 'RECORD ADDED', 'RECORD DELETED' or 'RECORDS UPDATED!!!!'.

6.1 Output Evidence

```
=====
***** WELCOME TO HIMALAYA WELLNESS *****
          SINCE -- 1930
=====
TODAY'S DATE  2024-11-23 15:20:36.606952
WE ARE OF TYPE PRIVATE
WE DEAL IN PRODUCTS:-
    CONSUMER GOODS
    HERBAL GOODS
    PERSONAL CARE
    CHILD CARE
=====
PRESS  1. TO DISPLAY PRODUCT RECORD FROM THE COMPANY'S STOCK
PRESS  2. TO ADD PRODUCT RECORD TO THE COMPANY'S STOCK
PRESS  3. TO DELETE PRODUCT RECORD FROM THE COMPANY'S STOCK
PRESS  4. TO UPDATE PRODUCT RECORD TO THE COMPANY'S STOCK
PRESS  5. TO SORT PRODUCT RECORD FROM THE COMPANY'S STOCK
PRESS  6. TO GIVE ORDER FOR PRODUCT
PRESS  7. TO SEE ORDER HISTORY
PRESS  8. TO SEE ORDER INVOICES
PRESS  9. TO CANCEL ORDER
PRESS 10. TO DELETE INVOICES
PRESS 11. TO EXIT
=====
SELECT YOUR CHOICE:-|
```

Figure 1. Main company screen showing the eleven available inventory and order operations.

```
=====
SELECT YOUR CHOICE:-1
=====
***** Welcome TO HIMALAYA COMPANY PRIVATE LIMITED *****
=====
-----YOU HAVE CHOSEN THE OPTION TO display PRODUCT'S DETAILS-----
PRESS 1 TO DISPLAY PRODUCTS INFO
PRESS 2 FOR EXIT
Enter Your Choice1
--NAME :NUMBER:IN STOCK:IN PRODUCTION:MRP; RATE, ; MANUFACTURE DATE, EXPIRY DATE --
|| ('NEEM FACEWASH', 10001, 10000, 5000, 120, 110, datetime.date(2024, 11, 1), datetime.date(2027, 10, 1)) ||
|| ('NEEM SCRUB', 10002, 11000, 5600, 118, 109, datetime.date(2024, 11, 9), datetime.date(2026, 9, 9)) ||
|| ('BABY SOAP', 10003, 12000, 4800, 65, 59, datetime.date(2024, 10, 10), datetime.date(2028, 10, 10)) ||
|| ('BABY POWDER', 10004, 12500, 5660, 110, 99, datetime.date(2024, 9, 8), datetime.date(2027, 9, 9)) ||
|| ('BABY CREAM', 10005, 13500, 7800, 75, 70, datetime.date(2024, 11, 11), datetime.date(2025, 9, 6)) ||

=====
***** Welcome TO HIMALAYA COMPANY PRIVATE LIMITED *****
=====
-----YOU HAVE CHOSEN THE OPTION TO display PRODUCT'S DETAILS-----
PRESS 1 TO DISPLAY PRODUCTS INFO
PRESS 2 FOR EXIT
Enter Your Choice2
```

Figure 2. Display option showing the product record fields and sample product records.

6.2 Output Evidence

```
=====
SELECT YOUR CHOICE:-2
=====
**** WELCOME TO HIMALAYA COMPANY PRIVATE LIMITED ****
=====
-----YOU HAVE CHOSEN THE OPTION TO ADD PRODUCT'S DETAILS-----
1. ADD PRODUCT
2. SHOW ADDED PRODUCTS
3. SEARCH PRODUCTS
4. EXIT
=====

Enter Your Choice1
ENTER PRODUCT NAME :WALNUT SCRUB
ENTER PRODUCT NUMBER :10008
ENTER NO. OF PRODUCTS INSTOCK :14500
ENTER NO. OF PRODUCTS ORDERED4500
ENTER PRODUCT MRP :135
ENTER PRODUCTS SELLING RATE :125
ENTER PRODUCT'S DATE OF MANUFACTURE : 2024-09-04
ENTER PRODUCT'S DATE OF EXPIRY: 2029-08-03
RECORDS ADDED
=====
```

Figure 3. Add-product menu and successful addition of a Walnut Scrub record.

```
=====
**** WELCOME TO HIMALAYA COMPANY PRIVATE LIMITED ****
=====
-----YOU HAVE CHOSEN THE OPTION TO ADD PRODUCT'S DETAILS-----
1. ADD PRODUCT
2. SHOW ADDED PRODUCTS
3. SEARCH PRODUCTS
4. EXIT
=====

Enter Your Choice2
NAME ;NUMBER;PRODUCTS NO. OF UNITS IN STOCK; PRODUCTS ORDERED;MRP; RATE, ; MANUFACTURE DATE, EXPIRY DATE

('NEEM FACEWASH', 10001, 10000, 5000, 120, 110, datetime.date(2024, 11, 1), datetime.date(2027, 10, 1))
('NEEM SCRUB', 10002, 11000, 5600, 118, 109, datetime.date(2024, 11, 9), datetime.date(2026, 9, 9))
('BABY SOAP', 10003, 12000, 4800, 65, 59, datetime.date(2024, 10, 10), datetime.date(2028, 10, 10))
('BABY POWDER', 10004, 12500, 5660, 110, 99, datetime.date(2024, 9, 8), datetime.date(2027, 9, 9))
('BABY CREAM', 10005, 13500, 7800, 75, 70, datetime.date(2024, 11, 11), datetime.date(2025, 9, 6))
('BODY LOTION', 10006, 16000, 7500, 135, 128, datetime.date(2024, 7, 7), datetime.date(2029, 8, 8))
('MOISTURISING LOTION', 10007, 12300, 3500, 145, 138, datetime.date(2024, 5, 5), datetime.date(2028, 9, 7))
('WALNUT SCRUB', 10008, 14500, 4500, 135, 125, datetime.date(2024, 9, 4), datetime.date(2029, 8, 3))
=====
```

Figure 4. Added-product display showing the stored product records.

6.3 Output Evidence

```
=====
SELECT YOUR CHOICE:-2
=====
**** WELCOME TO HIMALAYA COMPANY PRIVATE LIMITED ****
=====
-----YOU HAVE CHOSEN THE OPTION TO ADD PRODUCT'S DETAILS-----
1. ADD PRODUCT
2. SHOW ADDED PRODUCTS
3. SEARCH PRODUCTS
4. EXIT
=====
Enter Your Choice3
ENTER PRODUCT NUMBER: 10005
NAME , PRODUCT NUMBER; NO. OF UNITS IN STOCK; PRODUCTS ORDERED;MRP; RATE;MANUFACTURE DATE, EXPIRY DATE

[('BABY CREAM', 10005, 13500, 7800, 75, 70, datetime.date(2024, 11, 11), datetime.date(2025, 9, 6))]
```

Figure 5. Search operation using product number 10005 and the corresponding Baby Cream record.

```
=====
SELECT YOUR CHOICE:-3
=====
***** Welcome TO HIMALAYA COMPANY PRIVATE LIMITED *****
=====
-----YOU HAVE CHOSEN THE OPTION TO DELETE PRODUCT'S DETAILS -----
SELECT ONE OF THESE OPTIONS TO DELETE PRODUCT'S DETAILS
PRESS 1 USING PRODUCT NAME
PRESS 2 USING PRODUCT NUMBER
PRESS 3 FOR EXIT
Enter Your Choice :-1
ENTER PRODUCT NAME TO BE DELETED :- WALNUT SCRUB
[('WALNUT SCRUB', 10008, 14500, 4500, 135, 125, datetime.date(2024, 9, 4), datetime.date(2029, 8, 3))]
Is Being DELETED
RECORD DELETED
```

Figure 6. Delete operation using the product name Walnut Scrub.

```
=====
***** Welcome TO HIMALAYA COMPANY PRIVATE LIMITED *****
=====
-----YOU HAVE CHOSEN THE OPTION TO DELETE PRODUCT'S DETAILS -----
SELECT ONE OF THESE OPTIONS TO DELETE PRODUCT'S DETAILS
PRESS 1 USING PRODUCT NAME
PRESS 2 USING PRODUCT NUMBER
PRESS 3 FOR EXIT
Enter Your Choice :-2
ENTER PRODUCT NUMBER TO BE DELETED :- 10006
[('BODY LOTION', 10006, 16000, 7500, 135, 128, datetime.date(2024, 7, 7), datetime.date(2029, 8, 8))]
Is Being DELETED
RECORD DELETED
```

Figure 7. Delete operation using product number 10006.

6.4 Output Evidence

```

=====
SELECT YOUR CHOICE:-4
=====
***** Welcome TO HIMALAYA COMPANY PRIVATE LIMITED *****
=====
-----YOU HAVE CHOSEN THE OPTION TO UPDATE PRODUCT DETAILS -----
WHAT DO YOU WANT TO UPDATE
press 1 For Product Name
press 2 For Product No
press 3 For Product in Stock
press 4 For No of units Ordered
press 5 For MRP
press 6 For Selling Rate
press 7. EXIT
Enter Your Choice :- 1
ENTER PRODUCT NAME TO BE UPDATED: - NEEM SCRUB
[('NEEM SCRUB', 10002, 11000, 5600, 118, 109, datetime.date(2024, 11, 9), datetime.date(2026, 9, 9))]
ENTER PRODUCT NAME TO BE UPDATED WITH:-MOISTURISING LOTION
UPDATE DETAILS|| [('MOISTURISING LOTION', 10002, 11000, 5600, 118, 109, datetime.date(2024, 11, 9), datetime.date(2026, 9, 9))] ||
RECORDS UPDATED!!!!
=====

```

Figure 8. Update operation: changing the product name from Neem Scrub to Moisturising Lotion.

```

***** Welcome TO HIMALAYA COMPANY PRIVATE LIMITED *****
=====
-----YOU HAVE CHOSEN THE OPTION TO UPDATE PRODUCT DETAILS -----
WHAT DO YOU WANT TO UPDATE
press 1 For Product Name
press 2 For Product No
press 3 For Product in Stock
press 4 For No of units Ordered
press 5 For MRP
press 6 For Selling Rate
press 7. EXIT
Enter Your Choice :- 2
ENTER PRODUCT NAME WHOSE PRODUCT NUMBER IS TO BE UPDATED: -BABY CREAM
[('BABY CREAM', 10004, 13500, 7800, 75, 70, datetime.date(2024, 11, 11), datetime.date(2025, 9, 6))]
ENTER PRODUCT NUMBER TO BE UPDATED WITH: - 10005
UPDATE DETAILS|| [('BABY CREAM', 10005, 13500, 7800, 75, 70, datetime.date(2024, 11, 11), datetime.date(2025, 9, 6))] ||
RECORDS UPDATED!!!!
=====

```

Figure 9. Update operation: changing Baby Cream's product number.

```

***** Welcome TO HIMALAYA COMPANY PRIVATE LIMITED *****
=====
-----YOU HAVE CHOSEN THE OPTION TO UPDATE PRODUCT DETAILS -----
WHAT DO YOU WANT TO UPDATE
press 1 For Product Name
press 2 For Product No
press 3 For Product in Stock
press 4 For No of units Ordered
press 5 For MRP
press 6 For Selling Rate
press 7. EXIT
Enter Your Choice :- 3
ENTER PRODUCT NAME WHOSE TOTAL INSTOCK VALUE IS TO BE UPDATED:-NEEM FACEWASH
[('NEEM FACEWASH', 10001, 10000, 5000, 120, 110, datetime.date(2024, 11, 1), datetime.date(2027, 10, 1))]
ENTER THE NEW INSTOCK VALUE :-10500
UPDATE DETAILS || [('NEEM FACEWASH', 10001, 10500, 5000, 120, 110, datetime.date(2024, 11, 1), datetime.date(2027, 10, 1))] ||
RECORDS UPDATED!!!!
=====

```

Figure 10. Update operation: changing Neem Facewash stock quantity.

6.5 Output Evidence

```
***** Welcome TO HIMALAYA COMPANY PRIVATE LIMITED *****
-----YOU HAVE CHOSEN THE OPTION TO UPDATE PRODUCT DETAILS -----
WHAT DO YOU WANT TO UPDATE
press 1 For Product Name
press 2 For Product No
press 3 For Product in Stock
press 4 For No of units Ordered
press 5 For MRP
press 6 For Selling Rate
press 7. EXIT
Enter Your Choice :- 4
ENTER PRODUCT NAME WHOSE NO. OF UNITS ORDERED IS TO BE UPDATED :- NEEM FACEWASH
[(NEEM FACEWASH', 10001, 10500, 5000, 120, 110, datetime.date(2024, 11, 1), datetime.date(2027, 10, 1))]
ENTER THE NEW NO. OF UNITS ORDERED TO BE UPDATED WITH:-5500
UPDATE DETAILS || [(NEEM FACEWASH', 10001, 10500, 5500, 120, 110, datetime.date(2024, 11, 1), datetime.date(2027, 10, 1))] ||
RECORDS UPDATED!!!!
```

Figure 11. Update operation: changing the number of units ordered for Neem Facewash.

```
***** Welcome TO HIMALAYA COMPANY PRIVATE LIMITED *****
-----YOU HAVE CHOSEN THE OPTION TO UPDATE PRODUCT DETAILS -----
WHAT DO YOU WANT TO UPDATE
press 1 For Product Name
press 2 For Product No
press 3 For Product in Stock
press 4 For No of units Ordered
press 5 For MRP
press 6 For Selling Rate
press 7. EXIT
Enter Your Choice :- 5
ENTER PRODUCT NAME WHOSE MRP IS TO BE UPDATED NEEM FACEWASH
[(NEEM FACEWASH', 10001, 10500, 5500, 120, 110, datetime.date(2024, 11, 1), datetime.date(2027, 10, 1))]
ENTER THE NEW MRP OF THE PRODUCT 125
UPDATE DETAILS || [(NEEM FACEWASH', 10001, 10500, 5500, 125, 110, datetime.date(2024, 11, 1), datetime.date(2027, 10, 1))] ||
RECORDS UPDATED!!!!
```

Figure 12. Update operation: changing the MRP of Neem Facewash.

```
***** Welcome TO HIMALAYA COMPANY PRIVATE LIMITED *****
-----YOU HAVE CHOSEN THE OPTION TO UPDATE PRODUCT DETAILS -----
WHAT DO YOU WANT TO UPDATE
press 1 For Product Name
press 2 For Product No
press 3 For Product in Stock
press 4 For No of units Ordered
press 5 For MRP
press 6 For Selling Rate
press 7. EXIT
Enter Your Choice :- 6
ENTER PRODUCT NAME WHOSE SELLING RATE IS TO BE UPDATED :- NEEM FACEWASH
[(NEEM FACEWASH', 10001, 10500, 5500, 125, 110, datetime.date(2024, 11, 1), datetime.date(2027, 10, 1))]
ENTER THE NEW SELLING RATE :-115
UPDATE DETAILS || [(NEEM FACEWASH', 10001, 10500, 5500, 125, 115, datetime.date(2024, 11, 1), datetime.date(2027, 10, 1))] ||
RECORDS UPDATED!!!!
```

Figure 13. Update operation: changing the selling rate of Neem Facewash.

7. Testing and Observed Results

Test	Input / Operation	Observed result
Display	Menu option 1	Product records are displayed with their stored fields.
Add	Walnut Scrub and its details	The record is accepted and the output states that records were added.
Search	Product number 10005	Baby Cream record is displayed.
Delete by name	Walnut Scrub	The selected record is shown as being deleted and deletion is confirmed.
Delete by number	10006	Body Lotion record is identified and deletion is confirmed.
Update name	Neem Scrub → Moisturising Lotion	The updated record is displayed and update is confirmed.
Update product number	Baby Cream → 10005	The new product number appears in the updated record.
Update stock	Neem Facewash → 10500	The stock value changes and the record is updated.
Update ordered quantity	Neem Facewash → 5500	The ordered quantity changes and update is confirmed.
Update MRP	Neem Facewash → 125	The MRP changes and update is confirmed.
Update selling rate	Neem Facewash → 115	The selling rate changes and update is confirmed.

8. Conclusion

The supplied execution evidence demonstrates a functional console-based inventory management system for Himalaya Company Private Limited. The screenshots show that product records can be displayed, added, searched, deleted and updated through menu-driven interaction. The update demonstrations cover multiple individual fields, including product name, product number, stock, ordered quantity, MRP and selling rate. The consistent confirmation messages in the outputs provide visible evidence that the selected operations were completed.

Overall, the project provides a practical example of using Python programming concepts to build an interactive record-management application. The report is based on the supplied program-output screenshots and documents only the functionality visibly demonstrated in those screenshots.

9. Future Scope

- Add stronger input validation for invalid product numbers, quantities and prices.
- Improve the console interface with clearer formatting and error messages.
- Add authentication for different types of users.
- Provide printable or exportable inventory reports.
- Add additional reporting and analytics for stock and orders.